

Statistics for Health Data Science - Assessment 1

Student ID: 14210037

1. Please provide a summary of the data, with the aid of graphs and tables.

Solution:

import data into R

```
> d <- read.csv("SHDS_Assessment1_data.csv", header = TRUE)
```

```
> tab <- table(d$Diabetes_012)
```

```
> percent <- round(tab / sum(tab) * 100, 1)
```

```
> paste0(tab, " (", percent, "%)")
```

```
[1] "213703 (84.2%)" "39977 (15.8%)"
```

The patients without diabetes are more than 5 times compared with diabetes.

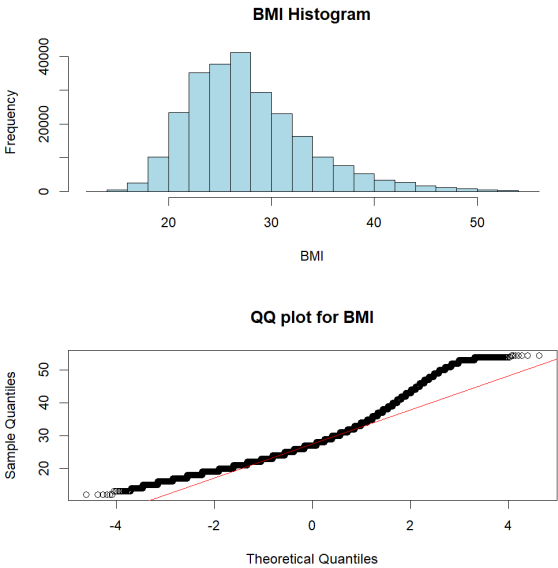
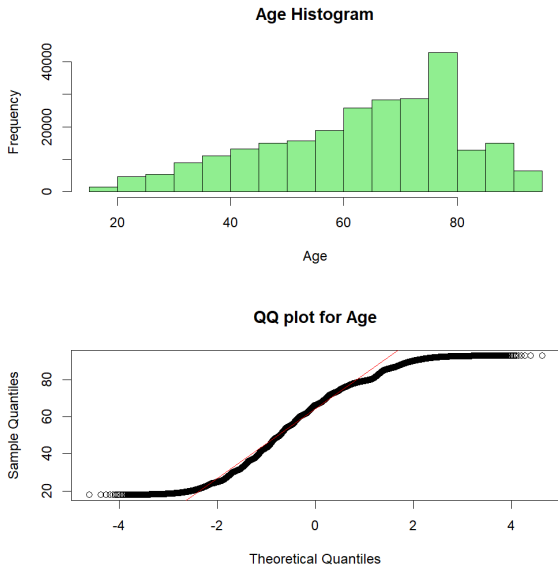
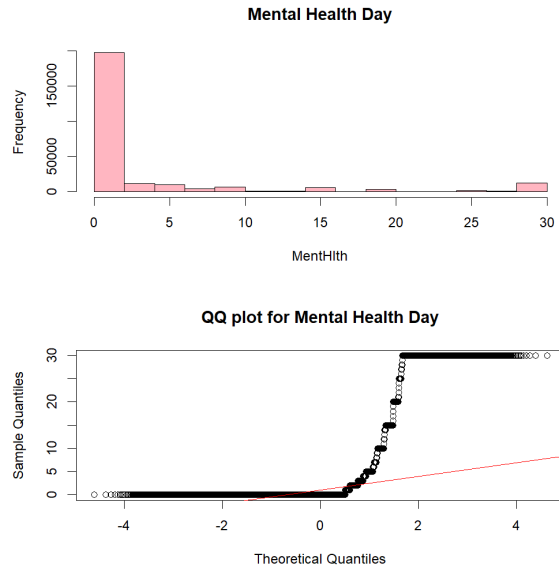
Table 1. Summary data from local organisation diabetes screening programme.

A. Internal comparison B. Compare the parameters with or without diabetes

Note: for Sex :0 = female, 1 = male

A	B.
<pre>> table(d\$HighChol)</pre> <pre> 0 1</pre> <pre>"146089 (57.6%)" "107591 (42.4%)"</pre>	<pre> HighChol Diabetes = 0 Diabetes = 1 </pre> <pre> :----- :----- :----- </pre> <pre> 0 132673 (90.8%) 13416 (9.2%) </pre> <pre> 1 81030 (75.3%) 26561 (24.7%) </pre>
<pre>> table(d\$CholCheck)</pre> <pre> 0 1</pre> <pre>"9470 (3.7%)" "244210 (96.3%)"</pre>	<pre> CholCheck Diabetes = 0 Diabetes = 1 </pre> <pre> :----- :----- :----- </pre> <pre> 0 9167 (96.8%) 303 (3.2%) </pre> <pre> 1 204536 (83.8%) 39674 (16.2%) </pre>
<pre>> table(d\$Smoker)</pre> <pre> 0 1</pre> <pre>"141257 (55.7%)" "112423 (44.3%)"</pre>	<pre> Smoker Diabetes = 0 Diabetes = 1 </pre> <pre> :----- :----- :----- </pre> <pre> 0 121879 (86.3%) 19378 (13.7%) </pre> <pre> 1 91824 (81.7%) 20599 (18.3%) </pre>
<pre>> table(d\$Stroke)</pre> <pre> 0 1</pre> <pre>"243388 (95.9%)" "10292 (4.1%)"</pre>	<pre> Stroke Diabetes = 0 Diabetes = 1 </pre> <pre> :----- :----- :----- </pre> <pre> 0 206944 (85%) 36444 (15%) </pre> <pre> 1 6759 (65.7%) 3533 (34.3%) </pre>
<pre>> table(d\$Sex)</pre> <pre> 0 1</pre> <pre>"141974 (56%)" "111706 (44%)"</pre>	<pre> Sex Diabetes = 0 Diabetes = 1 </pre> <pre> :--- :----- :----- </pre> <pre> 0 120959 (85.2%) 21015 (14.8%) </pre> <pre> 1 92744 (83%) 18962 (17%) </pre>

Table 2. The distribution of the Body mass index (BMI), age, and number of days in the past 30 days the patient says they have suffered with mental health (including stress, depression, etc). A. Internal comparison B. Compare the parameters with or without diabetes

	Body Mass Index (BMI)	Age	Days of Mental Health Issues
A	 Figure 1. Histogram and QQ plot for BMI distribution	 Figure 2. Histogram and QQ plot for Age distribution	 Figure 3. Histogram and QQ plot for Days of Mental Health Issues distribution
B	Diabetes_012 BMI_avg_sd <fct> <chr> 0 27.61 (5.6) 1 31.5 (6.51)	Diabetes_012 Age_avg_sd <fct> <chr> 0 61.81 (17.58) 1 70.58 (12.86)	Diabetes_012 MentHlth_avg_sd <fct> <chr> 0 2.94 (7.06) 1 4.47 (8.94)

Based on Table 1, this analysis revealed that up to 96.3% of individuals had conducted the cholesterol check. Although only past half of the overall population had high cholesterol, 24.7% population have high cholesterol with diabetes, compared with 9.2%, who had not found abnormal cholesterol measurement. Furthermore, the proportion of individuals with a history of stroke was relatively low at 4.1%; however, 34.3% of them had diabetes, which is notably higher than the 15% observed among those without experiencing the stroke. The overall distributions of smoking status and sex showed only minor differences between groups.

According to Table 2, the original distribution of BMI appears right-skewed, whereas age demonstrates a left-skewed. Most patients encountered Mental Health issues fewer than three days. None of the three variables follow a normal distribution, as indicated by QQ plots. However, if the sample size is large, the central portion of the BMI and age distributions may approximate normality according to the Central Limit Theorem (CLT). When comparing groups by diabetic status (Part B), patients with diabetes tended to have a higher mean BMI (31.5 kg/m²), were older on average (70.58 years), and reported more days of mental health issues (4.47 days) than those without diabetes.

In conclusion, the sample population trends illustrate that individuals with diabetes differ notably from those without this chronic condition. The former might have higher cholesterol level and BMI, stroke experience, older age, and more frequently suffer from mental problems.

2. Another screening programme took place last year in a different region and they found 14% of individuals had diabetes and the average BMI and age of those attending clinics were 28kg/m² and 55 years, respectively. They also found a significant relationship between smoking and diabetes. Investigate if your data support these results and provide possible explanations to your findings. (15 marks)

Solution:

The term Central Limit Theorem (CLT) refers to the mean of a large number of independent and identically distributed random variables from any distribution approximately follows a normal distribution. Because the samples in the dataset are more than 2000, it will almost certainly be a normal distribution. Furthermore, the confidence interval is set at 95%, implying that Type 1 error rate of 5% is accepted. In other words, while the fact is true, there is 5% probability of incorrectly rejecting the null hypothesis.

I. A diabetes prevalence is 14%, as shown in the screening programme conducted in another region last year.

To assess whether the diabetes prevalence in the current year is similar to that of the last year. The data is a nominal variable. Use the z-test to calculate a one-proportion test compared with the sample. Since the sample size is large, a proportion test was conducted. The approximately close result, without Yates' continuity correction, is preferred.

The null hypothesis posited that the regional prevalence of diabetes is equal to 14%. (H0: $p = 0.14$)

The alternative hypothesis proposed that the regional prevalence of diabetes is not equal to 14%. (H1: $p \neq 0.14$)

```
> prop.test(sum(d$Diabetes_012 == 1), nrow(d), p = 0.14, alternative = "two.sided",  
correct = FALSE, conf.level = 0.95)
```

1-sample proportions test without continuity correction

X-squared = 651.79, df = 1, p-value < 2.2e-16

95 percent confidence interval: 0.1561756 0.1590113

sample estimates: p 0.1575883

The calculated prevalence was around 15.76% (sample estimates $p = 0.1575883$) in this year, with a 95% confidence interval (95%CI) ranging from 15.62% to 15.90%

(0.1562–0.1590). The p-value is less than 0.05 ($p\text{-value} < 2.2 \times 10^{-16}$), and the 95%CI does not include the hypothesised value of 14%, the null hypothesis was rejected. There is a possibility that diabetes prevalence increases in the region compared to the previous.

II. Individuals' features, such as age and BMI, who joined this programme differ from those region or not.

To compare the mean BMI and age of participants is equal between different regions. The statistic is a one sample t-test. Before conducting the analysis, the criteria need to be verified: 1. The data follow a normal distribution. 2. The observations/individuals are independent. 3. The data are continuous. The first and second points are supported by CLT. Since the information was collected by individual visit, those requirements are sufficient.

The null hypothesis assumes that the mean BMI is equal to 28 kg/m². ($H_0: \mu = 28$)

The alternative hypothesis stated that the mean BMI is not equal to 28 kg/m². ($H_1: \mu \neq 28$)

```
> t.test(d$BMI, mu = 28, conf.level = 0.95)
```

One Sample t-test

t = 18.713, df = 253679, p-value < 2.2e-16

95 percent confidence interval: 28.19714 28.24326

sample estimates: mean of x 28.2202

Although the observation mean is close to 28 kg/m², the confidence interval is narrow (95%CI = 28.19 - 28.24.) and does not include the reference value. Apart from that, the p-value is significantly below 0.05, which indicates the null hypothesis was rejected.

The null hypothesis assumes that the mean age is equal to 55 years. ($H_0: \mu = 55$)

The alternative hypothesis executed that the mean age is not equal to 55 years. ($H_1: \mu \neq 55$)

```
> t.test(d$Age, mu = 55, conf.level = 0.95)
```

One Sample t-test

t = 239.53, df = 253679, p-value < 2.2e-16

95 percent confidence interval: 63.12380 63.25784

sample estimates: mean of x 63.19082

Because p-value is 2.2×10^{-16} and the mean age is 63.19 years (95%CI = 63.12 – 63.25), the alternative hypothesis was accepted. The mean age in this region is notably older than in the other areas.

III. Finding the correlation between smoking and diabetes.

The observation from the previous year found the association between smoking behaviour and diabetes prevalence. Due to the binary categorical data, the statistic included a Chi-squared test, as well as calculations of Relative Risk (RR), Odds ratio (OR), and Probability Difference (PD).

H0: no association between smoking and Diabetes

H1: an association between smoking and Diabetes

```
> chisq.test(d$Diabetes_012,d$Smoker,correct=FALSE)
```

Pearson's Chi-squared test

data: d\$Diabetes_012 and d\$Smoker

X-squared = 999.76, df = 1, p-value < 2.2×10^{-16}

```
> twoby2(table(Exposure = d$Smoker, Disease = d$Diabetes_012))
```

	0	1	P(0) 95% conf. interval		
0	121879	19378	0.8628	NA	NA
1	91824	20599	0.8168	0.8145	0.819
95% conf. interval					
Relative Risk: 1.0564				1.0527	1.0600
Sample Odds Ratio: 1.4109				1.3811	1.4415
Probability difference: 0.0460				0.0432	0.0489
Asymptotic P-value: 0.0000					

In this region, smokers had 1.05 times the risk of having diabetes compared with a non-smoker (95% CI: 1.05–1.06). According to the Sample Odds Ratio, the patient who is smoking has 1.41 times the odds (95% CI: 1.38 - 1.44) of having diabetes. the proportion of the diabetic group among smokers is 4.6% higher than non-smokers (95% CI: 0.043–0.048). The p-values are below the conventional threshold from both the chi-squared test and the contingency tables analysis, which supports the null hypothesis is rejected. These findings suggest a statistically significant correlation between smoking and diabetes. However, the possibility of a Type I error should be considered.

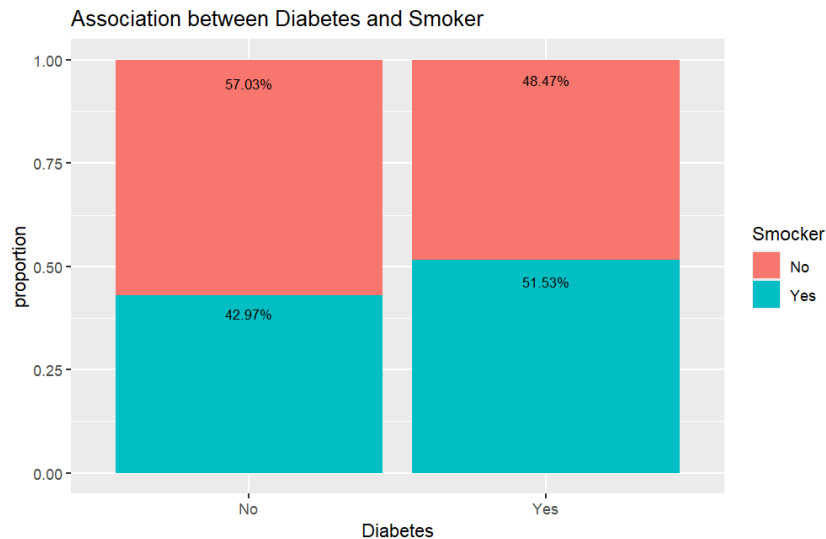


Figure 4. Proportion of smoking status stratified by diabetes diagnosis

According to the plot, 53.53% of individuals with diabetes were smokers, compared with 42.97% among non-diabetics. This visual pattern is aligned with the above findings and demonstrates a potential behavioural link to disease prevalence.

IV. Further investigations to support these results.

For further exploration, stratified analysis will be conducted to examine demographic co-factors influencing the relationship between smoking and diabetes. As indicated in Table 1, subgroup comparisons are appropriate when the distribution of individuals across grouping parameters is approximately equal. Therefore, this analysis will exclude variables with extreme values for the baseline characteristics. For instance, a large number of individuals have conducted cholesterol checks, while a smaller portion of the population has experienced a stroke. Additionally, data on mental health issues within the past thirty days exhibit severe skew.

The boundaries were defined in clinical terms. For BMI, thresholds 25, 30, and 40 kg/m² were used to represent overweight, obesity, and severe obesity, respectively. Age greater than 64-year-olds was classified as elderly.

Table 3. Stratified chi-squared tests to examine the association across key demographic variables.

Sex: female = 0, male =1	
> chisq.test(table(d\$Smoker[d\$Sex == 0], d\$Diabetes_012[d\$Sex == 0])) Pearson's Chi-squared test with Yates' continuity correction X-squared = 239.67, df = 1, p-value < 2.2e-16	> chisq.test(table(d\$Smoker[d\$Sex == 1], d\$Diabetes_012[d\$Sex == 1])) Pearson's Chi-squared test with Yates' continuity correction X-squared = 767.99, df = 1, p-value < 2.2e-16
High cholesterol: yes = 0, no = 1	
> chisq.test(table(d\$Smoker[d\$HighChol == 0], d\$Diabetes_012[d\$HighChol == 0])) Pearson's Chi-squared test with Yates' continuity correction X-squared = 259.81, df = 1, p-value < 2.2e-16	> chisq.test(table(d\$Smoker[d\$HighChol == 1], d\$Diabetes_012[d\$HighChol == 1])) Pearson's Chi-squared test with Yates' continuity correction X-squared = 253.9, df = 1, p-value < 2.2e-16
BMI (kg/m ²)	
25 > chisq.test(table(d\$Smoker[d\$BMI < 25], d\$Diabetes_012[d\$BMI < 25])) Pearson's Chi-squared test with Yates' continuity correction X-squared = 220.43, df = 1, p-value < 2.2e-16	> chisq.test(table(d\$Smoker[d\$BMI >= 25], d\$Diabetes_012[d\$BMI >= 25])) Pearson's Chi-squared test with Yates' continuity correction X-squared = 707.39, df = 1, p-value < 2.2e-16
30	X-squared = 540.73, df = 1, p-value < 2.2e-16
40	X-squared = 958.94, df = 1, p-value < 2.2e-16
	X-squared = 54.307, df = 1, p-value = 1.715e-13
Age: 65 years old	
> chisq.test(table(d\$Smoker[d\$Age < 65], d\$Diabetes_012[d\$Age < 65])) Pearson's Chi-squared test with Yates' continuity correction X-squared = 459.59, df = 1, p-value < 2.2e-16	> chisq.test(table(d\$Smoker[d\$Age >= 63.19], d\$Diabetes_012[d\$Age >= 63.19])) Pearson's Chi-squared test with Yates' continuity correction X-squared = 179.36, df = 1, p-value < 2.2e-16

Because all p-values are below the conventional threshold of 0.05, greater emphasis is placed on the chi-squared (χ^2) statistics to interpret the strength of association. While grouping individuals by having high cholesterol or not, the χ^2 are 259.81 and 253.9. The results suggest that the presence or absence of high cholesterol might not affect the relation between smoking and diabetes. In contrast, sex-based stratification reveals a stronger association in males, with a higher χ^2 value compared to females. A similar pattern is observed in the younger population. The results in BMI stratification are various. When using the overweight threshold as the cut-off, the χ^2 value is higher in the greater BMI group, suggesting a stronger association in individuals with unhealthier weight profiles. However, if the cut-off is shifted to reflect obesity or severe obesity, the statistical relationship reverses, indicating that the association may weaken in more extreme BMI categories.

To conclude, the association between smoking and diabetes remains statistically significant across all subgroups. Nevertheless, the strength of this association tends to be greater among males, younger individuals, and those with overweight BMI. In contrast, high cholesterol status appears to have minimal impact on this relationship.